

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

PHYSICS ORDINARY LEVEL

6118/3

PAPER 3

1 hour 15 minutes

Marks 40

2024

Additional Materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You may use a non-programmable calculator.
- Take the weight of 1 kg to be 10 N (i.e. acceleration of free fall $g = 10 \text{ m/s}^2$).

For Examiner's Use	
1	
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Marker	
Checker	

This document consists of **12** printed pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 A student investigates the effect of air resistance on the swing of a pendulum. She carries out the investigation in a series of steps.

Step 1

- (a) The student measures the mass m of the pendulum bob. Fig. 1.1 shows the reading on the balance.

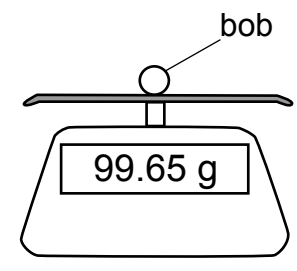


Fig. 1.1

Record mass m of the pendulum bob to the nearest gram.

$m = \dots\dots\dots$ g [1]

Step 2

- (b) The student sets up the pendulum as shown in Fig. 1.2.

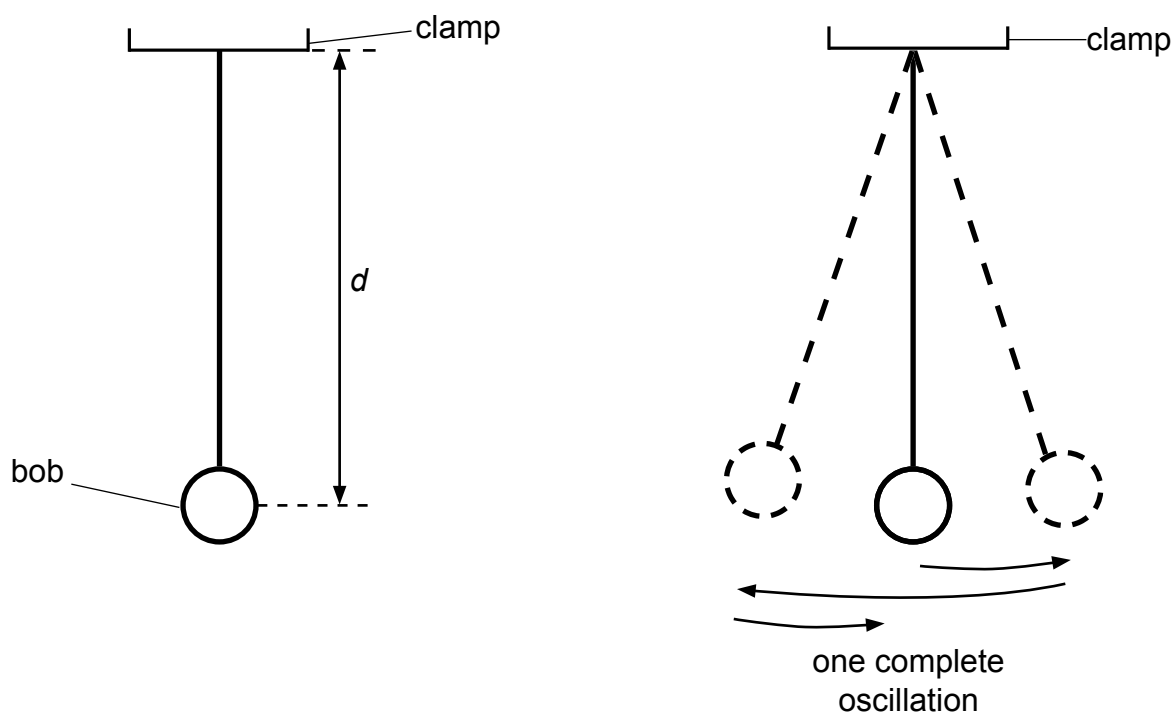


Fig. 1.2

She displaces the bob slightly and releases it so that it swings. She measures the time t_1 , t_2 and t_3 for 10 complete oscillations.

- (i) Record the times t_1 , t_2 and t_3 shown in Fig. 1.3.



Fig. 1.3

t_1 s

t_2 s

t_3 s [1]

- (ii) Calculate the average time taken t_{ave} for 10 oscillations.

$t_{\text{ave}} =$ s [1]

- (iii) Determine the period T of the pendulum.

The period of the pendulum is the time taken for one complete oscillation.

$T =$ s [1]

- (iv) State the advantage of repeating the time t readings three times.

..... [1]

Step 3

- (c) The student repeats **Step 2** by attaching a square sheet of a thick card to the pendulum to provide air resistance when the pendulum swings. She repeats the step using six different cards. Table 1.1 shows the results.

Table 1.1

size of card / cm × cm	T / s
5 × 5	1.526
6 × 6	1.532
7 × 7	1.512
8 × 8	1.540
9 × 9	1.553
10 × 10	1.542

- (i) Another student suggests that the size of the card is directly proportional to the period T .

State whether the results agree with this suggestion.
Justify your answer by making reference to the results.

statement

justification [2]

- (ii) Apart from the length d and mass m , state **one** other variable that should be kept constant in order to obtain reliable results in this experiment.

..... [1]

[8]

- 2 A student carried out an experiment to find the spring constant k of a steel spring. He sets up the apparatus as shown in Fig. 2.1.

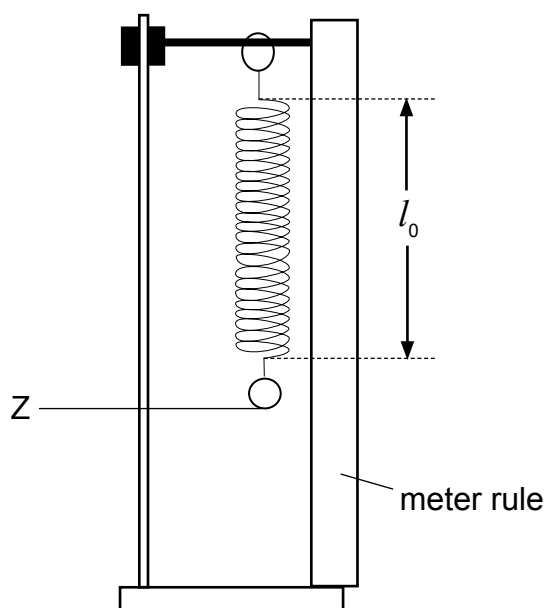


Fig. 2.1

The student measures the unstretched length l_0 of the spring, in mm on Fig. 2.1 and records it as shown.

$$l_0 = \dots\dots\dots 39 \dots\dots\dots \text{mm}$$

- (a) Explain why l_0 is not measured to point Z on the spring.

.....

.....

[1]

(b) The student hangs a 1.0 N load L to the spring.

He records the new length l of the spring.

He repeats the procedure using loads $L = 2.0$ N, $L = 3.0$ N, $L = 4.0$ N, $L = 5.0$ N and $L = 6.0$ N.

The readings are shown in Table 2.1.

Table 2.1

L / N	l / mm	e / mm
1.0	42	
2.0	43	
3.0	46	
4.0	49	
5.0	50	
6.0	52	

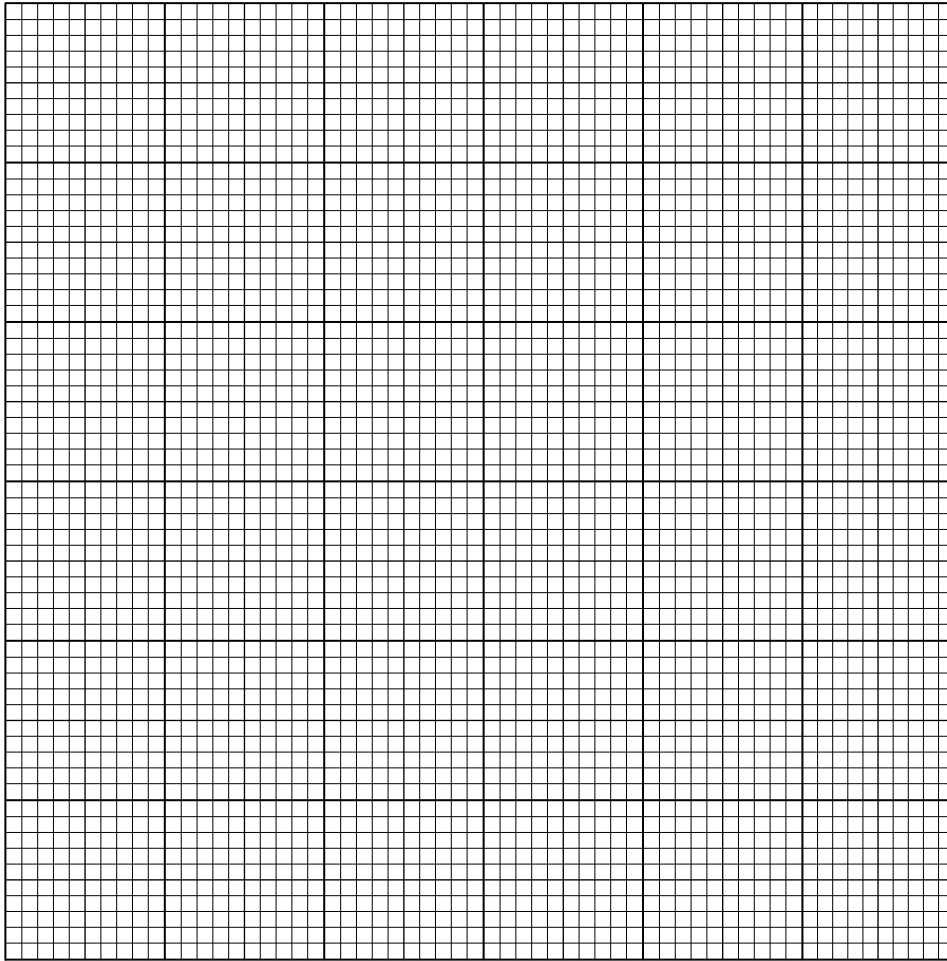
Calculate the extension e of the spring produced by each load, using the equation.

$$e = (l - l_0).$$

Record the values of e in Table 2.1.

[2]

- (c) Use Table 2.1 to plot a graph of extension e on the y -axis and load L on the x -axis. Draw a line of best fit.



[5]

- (d) Theory suggests that the gradient of the graph is equal to the spring constant k of the spring.

Use the graph to determine k . Show clearly how you obtained the necessary information to calculate the gradient.

Include the unit.

gradient =

unit =

[4]

[12]

- 3 A student investigates how the resistance of a wire varies with length as shown in Fig. 3.1.

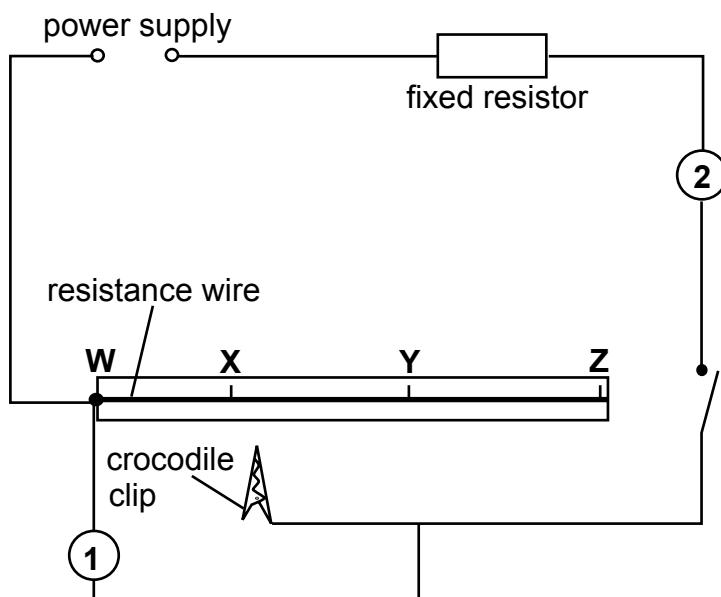


Fig. 3.1

- (a) State the names of instruments **1** and **2** shown in Fig. 3.1 that are used to measure potential difference across and the current through the resistance wire.

1

2

[2]

- (b) The student connects the crocodile clip to point **Z**, where **WZ** is of length $l = 90$ cm of the resistance wire. The student measures the potential difference V across **WZ** and the current I in the circuit.

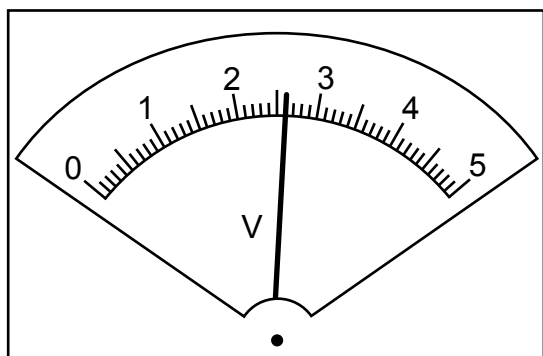


Fig. 3.2

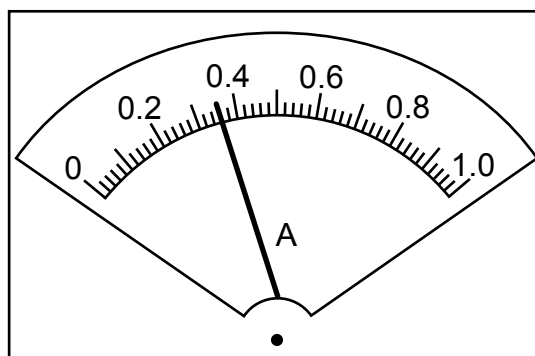


Fig. 3.3

Record the potential difference V and current I .

$V =$

$I =$

[2]

- (c) The student then connects the crocodile clip to lengths **WY** of length $l = 60$ cm and **WX** $l = 40$ cm. The corresponding potential differences V across the resistance wire are also measured.

The readings are shown in Table 3.1.

Table 3.1

l / cm	$V / \dots\dots\dots$	$I / \dots\dots\dots$	R / Ω	$\frac{R}{l} / \frac{\Omega}{\text{cm}}$
60.0	2.5	0.52		
40.0	2.3	0.71		

- (i) Complete the column headings in Table 3.1. [1]
- (ii) Calculate, and record in Table 3.1, the resistance R of each length l of the wire. Use the readings from Table 3.1 and the equation $R = \frac{V}{I}$. [2]
- (iii) Calculate, and record in Table 3.1, the value of $\frac{R}{l}$ for each length of wire. [1]
- (d) Use your results in Table 3.1 to calculate the resistance R_{30} of a 30.0 cm length of the resistance wire.

Show your working.

$$R_{30} = \dots\dots\dots \Omega \quad [1]$$

- (e) Suggest **one** reason why different students, carrying out the experiment carefully with the same equipment may not get identical results.

.....

.....

..... [1]

- (f) The student finds that, during the experiment, the wire becomes hot because of a high current. She decides to use a variable resistor to prevent this. Complete the circuit in Fig. 3.4 to show a variable resistor used for this purpose in the experiment.

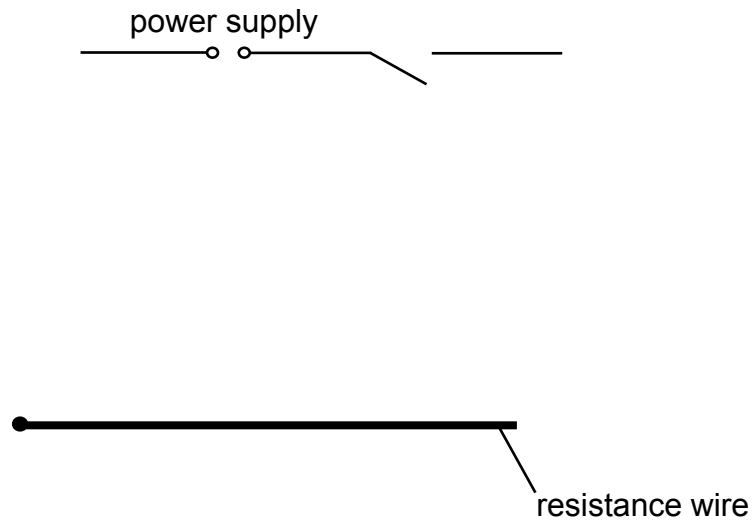


Fig. 3.4

[2]

[12]

- 4 A Physicist suggests that the volume of hot water will affect the rate of cooling of hot water in a container.

Plan an experiment to investigate the relationship between the volume of hot water and the rate of cooling.

Write a plan for the experiment, including:

- a brief description of how the investigation would be carried out including the apparatus that you would use
- which measurements should be taken, and how you will ensure that the results are accurate
- how the results would be processed
- the graph that would be plotted from the measurements taken - you should sketch the axes, with appropriate labels (you are not required to enter any readings in your graph)
- how a conclusion is reached

A diagram is not required but you may draw one if it helps to explain your plan.

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[8]

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